

Shea Cardozo

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RESEARCH INTERESTS

I am broadly interested in creating embodied autonomous agents and systems. My PhD work focuses chiefly on motion prediction and planning tasks for self-driving vehicles, and especially how to tackle the problems of covariate shift and casual confusion that hamper closed-loop performance of popular methods.

My current research direction is in hybrid rule-based and deep learning planners and especially the incorporation of code-generation models in human-in-the-loop rule engines. Other areas of interest of mine not directly related to my current work are model-based reinforcement learning, causal representation learning, and mechanistic interpretability techniques.

EDUCATION

2023 - 2027	PhD in Computer Science Waterloo Intelligence Systems Lab Supervisor: Professor Krzysztof Czarnecki	University of Waterloo
2021 - 2022	Master of Science in Applied Computing Data Science Concentration	University of Toronto
2017 - 2021	Bachelor of Mathematics in Statistics Co-operative Program. Dean's Honours.	University of Waterloo

SKILLS

Programming Languages	Proficiency with Python, R, and C++. Experience with Java, C#, DART, Haskell, Scheme (Racket), SQL, VBA, HTML/CSS.
Deep Learning Frameworks	Proficiency with PyTorch. Experience with Jax, TensorFlow
Development and Deployment	Git, Linux, AWS, GCP, VSCode, RStudio, Jupyter Notebooks

PUBLICATIONS

- Shea Cardozo, Gabriel Islas Montero, et al. 2022. *Explainer Divergence Scores (EDS): Some Post-Hoc Explanations May be Effective for Detecting Unknown Spurious Correlations*. Presented at the AIMLAI workshop at CIKM 2022. Available at [link](#).
- Eli-Henry Dykhne, Shea Cardozo, James Riddell, Sean Sedwards, Krzysztof Czarnecki. 2024. *Systematic Generation of Dynamic Occlusion-Caused Collisions*. Manuscript; work extended in E.-H. Dykhne, M.Sc. thesis, University of Waterloo, 2024. Available at [link](#).

AWARDS

- **University of Waterloo Graduate Scholarship** - \$5,000/year for one year Fall 2023
- **David R. Cheriton Graduate Scholarship** - \$10,000/year for two years Winter 2024

INDUSTRY EXPERIENCE

Machine Learning Scientist

May 2022 - Sept. 2023

Tenyks

- Chief Machine Learning Scientist at Tenyks (YC'S21) reporting directly to the CTO, researching and prototyping new methods for classification, detection, and segmentation tasks. Projects include:
 - Created a synthetic data pipeline for zero-shot anomaly detection via inserting photorealistic synthetic anomalies into clean data using RGB pixel masks and neural image inpainting.
 - Implemented a scalable system for open-set object retrieval based on visual and text prompting using 'Segment Anything' proposals and 'OpenCLIP' data embeddings. Fine-tuned 'OpenCLIP' embeddings on domain-specific data to improve retrieval performance beyond baseline.
 - Prototyped multiple active learning and core set approaches to suggest an efficient data annotation strategy for settings with minimal annotated data and expensive annotation costs.
 - Implemented a sequence of automated 'data quality checks' to detect common dataset issues in computer vision such as a high prevalence of occlusion or inconsistent class definitions.
 - Formulated a novel evaluation criteria for post-hoc neural network explanations to detect dependence on spurious correlations. Verified and submitted our work as a workshop paper.

Data Scientist - Claims AI Team

Sept. 2020 - Dec. 2020

Intact Insurance

- Constructed a pipeline to automatically classify insurance documents from image and text data using an ensembled 'ResNet' convolutional neural network and 'BERT' transformer neural network.
- Experimented with multi-objective non-gradient optimization methods such as the 'NSGA-II' genetic algorithm to optimize model prediction thresholds to mark unclassified documents for manual review.

Data Scientist - Analytics

May 2020 - Aug. 2020

Noom Inc.

- Specified and fit an autoregressive time series model with seasonal effects to predict the influx of user support tickets to ensure sufficient resource availability.
- Trained and benchmarked a set of 'GloVe' vector embeddings constructed from internal food data to improve user meal recommendation and tracking.

Actuarial Analyst - DataLab Division

Sept. 2019 - Dec. 2019

Intact Insurance

- As part of the 'Rating Revolution' team, trained 'XGBoost' gradient-boosted decision tree models to replace the existing generalized linear models used in home insurance pricing
- Created a Python visualization tool to analyze how different pricing models impact wider financials.

Associate Actuarial Programmer

Jan. 2019 - Apr. 2019,

Moodys Analytics

May 2018 - Aug. 2018

- Implemented highly performance sensitive financial calculations into our insurance software platform using the C++ programming language, with focus on long-term maintainability.
- Expanded UI functionality to more transparently display to clients how financials are calculated.

PERSONAL PROJECTS

Graddy



- Created a basic deep learning framework entirely from scratch in Python, including an implementation of AutoGrad full capable of handling the specification, training and inference of neural networks.
- Implemented data structures for generic modules and tensor computation, popular deep learning layers and activation functions, and optimization algorithms such as SGD and SGD with Momentum.

Adversarial Conditional UNET



- Trained a UNET model to generate adversarial examples using the CIFAR-10 dataset to fool a variety of state-of-the-art image classification models, inspired by existing work in adversarial denoising.
- Verified trained model can conduct targeted attacks on models not used in training with only marginal decrease in success rate, indicating successful generalization of the generated adversarial examples.

This JoJo Does Not Exist



- Trained a StyleGAN2 image generation model to generate faces that resemble characters from the manga "JoJo's Bizarre Adventure", using a custom dataset scraped using a 'Selenium' Python bot.
- Employed a Google Cloud Platform VM to train model for 72 hours on a Linux GPU instance.

Trump Tweet Generator



- Trained the GPT-2 338M language model on former US President Donald Trump's twitter feed.
- Built a web app using the 'Flask' Python package to display generated tweets.